Architectural Reconstruction of the Stela Governance System: The Pointer-First Constitution and the Specification of DP-OPS-0046

1. Executive Summary and Theoretical Framework

The evolution of repository-resident artificial intelligence necessitates a fundamental shift in governance architecture. Traditional models of "instruction-following" for Large Language Models (LLMs) rely on verbose, inline directives that consume valuable context window space and introduce susceptibility to "Semantic Drift"—a phenomenon where duplicated rules diverge over time, creating conflicting truth sources. This report analyzes the "Stela" platform's transition to a **Pointer-First Constitution**, a governance model that replaces "Ghost Canon" (duplicated text) with executable pointers to a Single Source of Truth (SSOT).

Forensic analysis of the repository state at **HEAD 46a04806** [1] reveals a critical inflection point. While previous Dispatch Packets (specifically DP-OPS-0045, recorded in the State of Play ledger [1]) initiated the reconstruction of the TASK.md surface, the system currently resides in a "Dirty State" with uncommitted changes, indicating an incomplete transition to the new constitutional standard. The rigorous application of the **Policy of Truth (PoT)** requires a new Dispatch Packet, **DP-OPS-0046**, to finalize this architecture, enforce the "Freshness Gate" protocols, and codify the "Receipt-Based" verification methodology.

The immediate imperative is to draft DP-OPS-0046. However, to execute this task with the "Nuanced Understanding" required by the Operator, one must first dissect the intricate web of dependencies, enforcement scripts, and constitutional axioms that define the Stela environment. This report provides that exhaustive technical analysis, culminating in the synthesized Dispatch Packet required to bring the system into equilibrium.

1.1 The Context Window Dilemma

In stateless AI interaction models, the "Context Window" is the scarcest resource. Every token dedicated to governance boilerplate—such as repeating behavioral rules or formatting instructions—is a token subtracted from the capacity for complex reasoning and payload execution.

Analysis of the Stela repository [1] indicates a sophisticated recognition of this bottleneck. The structure of the ops/ directory, specifically the existence of ops/bin/llms and ops/lib/manifests/CONTEXT.md, demonstrates an architectural intent to optimize context

loading. However, legacy patterns often persist where instructions are copied directly into active work surfaces like TASK.md. This creates "Ghost Canon": phantom rules that exist outside the authorized PoT.md but exert influence over the agent's behavior.

The "Pointer-First" methodology addresses this by treating governance documents (PoT.md, docs/MANUAL.md) not as text to be copied, but as libraries to be referenced. Just as a software function calls a library rather than pasting its source code, the Stela agent is engineered to "load" governance logic via pointers (e.g., "See PoT.md Section 4.2"). This resolves the Context Window Dilemma by compressing kilobytes of regulatory text into bytes of reference tokens.

1.2 Semantic Drift and the Single Source of Truth

A core axiom of the Stela Constitution [1] is the **Single Source of Truth (SSOT)**. When governance logic is duplicated—for example, if the "Linguistic Precision" standard is defined in PoT.md but also summarized in TASK.md—a divergence risk emerges. If the Operator updates the PoT but fails to update the summary in the Task dashboard, the agent encounters conflicting imperatives. This is "Semantic Drift."

The forensic state of TASK.md at HEAD 46a04806 suggests the repository is moving toward a model where TASK.md contains *no* normative rules, only structural scaffolding that points to the PoT. This ensures that any update to the Constitution propagates instantly to all active agents without requiring a refactor of the task templates.

1.3 The "Vibes-Based" Reporting Hazard

Stochastic AI models have a propensity for "Vibes-Based Reporting"—providing qualitative, subjective summaries of work (e.g., "I checked the code and it looks good") rather than quantitative evidence. The Stela architecture explicitly rejects this through its "Receipt" requirement.

Governance in Stela is binary: pass or fail. There is no partial credit. The analysis of tools/lint/truth.sh and tools/lint/style.sh [1] reveals a system designed to enforce "Physical Laws" on the code. For an agent to claim a task is complete, it must provide a "Receipt" containing the raw standard output (stdout) of these verification tools. DP-OPS-0046 is critical in codifying this requirement within the active contract.

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2. Forensic Analysis of the Stela Repository Architecture

To draft a compliant Dispatch Packet, one must first understand the terrain. The Stela repository is not merely a collection of files; it is a "Cybernetic Organism" where file paths

dictate jurisdiction and behavior. The "Filing Doctrine" defined in PoT.md Section 1.1 [1] establishes the fundamental physics of this universe.

2.1 The Four Domains of Jurisdiction

Domain	Path	Function	Constitutional Status
Run	ops/	Executable logic, binaries, manifests.	The "Kernel" of the system.
Explain	docs/	Manuals, rationale, maps.	Narrative justification, referenced by pointers.
Work	projects/	Payload code.	The subject of work, but never the governor.
Trash	storage/	Local artifacts, dumps, handoffs.	Ephemeral state, never Canon.

Insight: The strict separation of ops/ (Run) and docs/ (Explain) is critical. A common failure mode in repository management is "Documentation Drift," where the manual says one thing (docs/) but the script does another (ops/). Stela resolves this by making the ops/ directory the authority for *execution*. The docs/ directory explains *why*, but ops/ dictates *how*.

2.2 The Ops Kernel (ops/)

The ops/ directory is the most active surface in the repository manifest.[1] It contains the binaries that enforce the PoT.

- **ops/bin/open:** This binary generates the "Freshness Gate." It captures the git state (HEAD, branch, porcelain) and injects it into the prompt. This prevents the "Time Travel" hallucination where an AI assumes it is working on the latest code when it is actually looking at a stale context.
- **ops/bin/dump:** The "Observer" mechanism. It serializes the repository state into a format ingestible by the LLM. It supports "Scopes" (platform, full, project) to manage token density.
- **ops/bin/llms:** The "Context Bundler." This binary is vital for the "Flat Truth" architecture. It aggregates the scattered documentation into cohesive text files (llms.txt, llms-ops.txt)

at the repository root. This ensures that the AI has a "map" of the truth that is generated programmatically, rather than manually maintained.
- **ops/bin/map:** The "Wayfinding Maintainer." It updates docs/MAP.md automatically, ensuring that the human-readable map of the repo matches the actual file structure.

Forensic Note: The user query includes an addendum requiring the allowlisting of llms*.txt files if the llms command is invoked. This confirms that the Operator views the llms binary as a state-modifying tool that requires explicit permission to write to the repository root.

2.3 The Immune System (tools/lint/)

The tools/ directory functions as the repository's immune system, identifying and rejecting "foreign bodies" (drift, contractions, context hazards).

- **tools/lint/style.sh:** Enforces the "Linguistic Precision" clause of the PoT. It greps for contractions (e.g., "don't," "can't"). If found, the build fails. This forces the AI to speak in the "Stela Voice"—precise, rhythmic, and robotic.
- **tools/lint/truth.sh:** Enforces the "Policy of Truth." It checks for forbidden terms and ensures that Canon files (PoT.md, TASK.md) are present and correctly named.
- **tools/lint/context.sh:** Enforces "Context Hygiene." It verifies that ops/lib/manifests/CONTEXT.md does not include forbidden directories (like docs/library/agents). This prevents "Context Poisoning," where the AI floods its own memory with irrelevant instructions.
- **tools/lint/dp.sh:** The "Contract Validator." It checks the TASK.md file against a rigid schema. It ensures that the Dispatch Packet contains the required decimal headings (3.1, 3.2, etc.) and that the allowlist is present.

2.4 The Canon Surfaces

The "Canon" represents the immutable laws of the system.

- **PoT.md (Policy of Truth):** The Constitution. It defines the staffing roles (Operator, Integrator, Contractor) and the behavioral standards. It is the final authority in any conflict.
- **SoP.md (State of Play):** The Ledger. It records the history of what shipped and why. It is an append-only log of truth.
- **TASK.md (The Contract):** The Active Work Surface. It contains the current objective. It is the *only* file allowed to be in a "Dirty State" during session initialization.[1]

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3. Forensic State Assessment (HEAD 46a04806)

To draft DP-OPS-0046 effectively, we must understand the precise coordinates of the repository in spacetime. We rely on the OPEN artifact [1] and the SoP ledger.[1]

3.1 The Snapshot

- **Active Branch:** main.
- **HEAD Hash:** 46a04806.
- **Porcelain State:** "Dirty". The OPEN prompt [1] explicitly states: Porcelain (git status --porcelain): M TASK.md.

This "Dirty State" is highly significant. In most CI/CD pipelines, a dirty state is a failure condition. However, Stela's PoT.md [1] acknowledges a "Dirty State Exception" for TASK.md. This file serves as the "RAM" of the system—it holds the session state, the active pointers, and the work log. It *must* be mutable to record the progress of the agent. The presence of uncommitted changes in TASK.md indicates that a session is either in progress or was left incomplete.

3.2 The Historical Trajectory (SoP Analysis)

The SoP.md file [1] provides the narrative arc leading to this moment:

- **2026-02-11 01:52:19 UTC — DP-OPS-0045:** "Reconstructing and Perfecting TASK.md."
  - *Objective:* Reconstructed the template, restored Session State and Logic Pointers, and hardened the Freshness Gate.
  - *Result:* The template structure was updated, but the "Dirty State" at 46a04806 suggests that the final "seal" of this reconstruction—the update to the Constitution to formally recognize this structure—is pending.
- **2026-02-10 21:18:35 UTC — DP-OPS-0044:** "TASK Contract Hardening."
  - *Objective:* Enforced Base HEAD alignment for gate artifacts.
- **2026-02-10 18:03:22 UTC — DP-OPS-0043:** "Task Subsystem Hardening."

The trajectory is a recursive hardening process. The system is moving from a "loose" collection of markdown files to a "tight," programmatically enforced governance engine. DP-OPS-0046 is the next logical step: finalizing the "Pointer-First" constitution by removing any remaining duplicated text (Ghost Canon) that might have survived the DP-OPS-0045 reconstruction.

3.3 The "Addendum" Requirement

The user query contains a critical constraint: **"If the DP plan includes an llms command invocation or a context bundle refresh objective, the DP allowlist must include the repository root outputs: llms.txt, llms-small.txt, llms-full.txt, llms-ops.txt, and llms-governance.txt."**

Why is this relevant?

The objective of DP-OPS-0046 is "Pointer-first agent constitution." This implies modifying PoT.md or TASK.md. Since these files are part of the "Context Bundles" generated by

ops/bin/llms (specifically llms-governance.txt and llms-small.txt), any change to them *requires* a regeneration of the bundles to maintain the "Flat Truth." If we update the Constitution but fail to update the bundle, the llms.txt file (used for discovery) becomes stale. Therefore, DP-OPS-0046 *must* invoke ops/bin/llms, and consequently, the allowlist *must* include the root text files.

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4. The Pointer-First Paradigm: Theory and Implementation

The core objective of DP-OPS-0046 is to implement the **Pointer-First Agent Constitution**. This is a specific design pattern for AI governance.

4.1 The Problem: Ghost Canon

In the early iterations of Stela (visible in the legacy TASK.md patterns implied by the "Refinement" document [1]), instructions were likely copied into the task file.

- *Example:* "Behavioral Standard: Do not use contractions."
- *Risk:* If PoT.md changes to "Do not use contractions or emojis," the legacy TASK.md is now out of sync. This is "Semantic Drift."

4.2 The Solution: Executable Pointers

The Pointer-First model replaces the text with a reference.

- *New Pattern:* "Behavioral Standard: See PoT.md Section 4.2."
- *Mechanism:* When the AI reads "See PoT.md," it uses its retrieval capability (or the provided context dump) to look up the *current* truth at that location.
- *Benefit:* The TASK.md becomes immutable logic. It does not need to change when the policy changes. It simply points to the live policy.

4.3 Cognitive Load and Token Density

The "TASK.md Analysis" document [1] highlights the "Context Window Dilemma." A Pointer-First Constitution significantly reduces the token count of the TASK.md file. Instead of occupying 500 tokens with repeated rules, it uses 50 tokens for pointers. This frees up 450 tokens for "Payload Execution"—actual coding work.

Furthermore, the document notes a "Hybrid-Hardened Structural Design".[1] The pointers serve as "Cognitive Stop Signs." They force the model to flush its local attention cache and perform a lookup, which reduces the likelihood of "Instruction Bleed" (where instructions from a previous task contaminate the current one).

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5. Implementation Strategy for DP-OPS-0046

We are now prepared to construct the Dispatch Packet. The process follows the "A-E Canon" execution plan mandated by the Stela architecture.

5.1 Defining the Session State

The OPEN artifact [1] gives us our anchors:

- **Base Branch:** main.
- **Base HEAD:** 46a04806.
- **Dump:** storage/dumps/dump-full-main-46a04806.txt.

The DP must explicitly reference these. If the agent attempts to run this DP against a different HEAD (e.g., 46a04807), the "Freshness Gate" logic in ops/bin/open will fail, preserving integrity.

5.2 The "Logic Pointers" Block

This block effectively replaces the "Ghost Canon." Instead of listing the "Staffing Protocol," we list the pointer: Staffing Protocol: PoT.md Section 4.1.

We must also point to the toolchain:

- Linguistic Precision: tools/lint/style.sh.
- Structure Verification: tools/verify.sh.

This tells the agent: "The law is not what I say; the law is what this script enforces."

5.3 The Allowlist (The Firewall)

The allowlist is the security boundary. For DP-OPS-0046, the scope is:

1. **Refine TASK.md:** We must edit TASK.md.
2. **Update History:** We must edit SoP.md.
3. **Capture Proof:** We must write to storage/handoff/DP-OPS-0046-RESULTS.md.
4. **Refresh Context:** As discussed in Section 3.3, we must regenerate the llms bundles. This requires write access to:
    - llms.txt
    - llms-small.txt
    - llms-full.txt
    - llms-ops.txt
    - llms-governance.txt

Any attempt to edit a file *not* on this list (e.g., README.md or projects/code.php) will trigger a

"Drift" failure in the audit phase.

5.4 The Execution Logic

The execution plan must be strictly sequential:

1. **Pre-flight:** Run tools/lint/context.sh to ensure the environment is safe.
2. **Refine:** Edit TASK.md to remove duplicated text and ensure pointers are correct.
3. **Refresh:** Run ops/bin/llms to propagate the changes to the context bundles.
4. **Verify:** Run the full linter suite (truth, style, dp).
5. **Receipt:** Capture the output of these commands into the RESULTS.md file. *This is non-negotiable.* "Vibes-based" confirmation is not accepted.

5.5 Closeout and Transition

The DP must define the "Thread Transition." This involves appending a THREAD END marker to the work log. This marker signals to the "Prune" binary (ops/bin/prune, described in [1]) that the session is eligible for archival.

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6. The Dispatch Packet Construction (DP-OPS-0046)

The following artifact represents the synthesized command structure for the Stela Integrator. It is constructed to be "executable" by the Contractor (Guest AI) with zero ambiguity.

Key Structural Elements:

- **No Contractions:** "Do not" instead of "Don't."
- **Quantitative Reporting:** "Execution path deviated by X%" required in case of failure.
- **Decimal Indexing:** Sections are numbered 1, 2, 2.1, 3, 3.1 to support JSON-like parsing by the model.
- **Explicit Constraints:** The "Hard Constraints" section lists the SSOT files that cannot be ignored.

The resulting output adheres to the TASK.md template v2.0 as reconstructed in DP-OPS-0045, but refines it further to the "Pointer-First" standard.

6.1 Table of System Failure States (Drift Triggers)

Before drafting the DP, we review the failure states defined in PoT.md [1] to ensure the DP guards against them.

Failure Type	Definition	DP-OPS-0046 Mitigation

Canon Drift	Divergence between PoT.md and repo state.	The DP explicitly invokes tools/lint/truth.sh.
Context Hazard	Including library/ docs in CONTEXT.md.	The DP invokes tools/lint/context.sh as a gate.
Dirty State	Uncommitted changes in main.	The DP requires OPEN-PORCELAIN verification.
Ambiguity	Interpretations capable of multiple meanings.	The DP uses rigid "Absolute Literalism" syntax.

6.2 Table of Allowed Artifacts (Allowlist)

File	Justification
TASK.md	The subject of the refinement work.
SoP.md	The immutable ledger requiring an update entry.
storage/handoff/*	The location for the mandatory Results Receipt.
llms*.txt	The context bundles requiring regeneration (Addendum rule).

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7. Conclusion

The transition to a Pointer-First Constitution represents a maturation of the Stela platform from an experimental repository to a rigorously governed "Cybernetic Organism." By encoding the "Policy of Truth" into the file system and enforcing it via deterministic binaries (ops/bin/*), Stela eliminates the ambiguity that plagues standard Large Language Model integrations.

DP-OPS-0046 is the keystone of this transition. It locks the TASK.md surface into its final, pointer-based configuration, ensures that the context bundles (llms.txt) reflect this new

reality, and establishes the precedent that *no work occurs without a verified receipt*.

The following Dispatch Packet is the synthesized output of this architectural analysis, ready for immediate deployment by the Operator.

STELA TASK DASHBOARD (LIVING SURFACE)

Status: ACTIVE

Owner: Integrator

Last Updated: 2026-02-10

1. Session State (The Anchor)

Pointer: storage/handoff/OPEN-*.txt (The generated session context)

Active Branch: main (Must match OPEN artifact)

Base HEAD: 46a04806 (Must match OPEN artifact)

Context Manifest: ops/lib/manifests/CONTEXT.md (Checked by tools/lint/context.sh)

2. Logic Pointers (The Law)

Primary Constraint: PoT.md (Policy of Truth) wins in all conflicts.

2.1 Governance Pointers

Jurisdiction: PoT.md Section 3 (Ops, Docs, Projects, Tools).

Git Authority: PoT.md Section 4.1 (Operator owns main; Integrator owns TASK.md).

Behavioral Standard: PoT.md Section 4.2 (Linguistic Precision, Literalism).

Staffing Protocol: PoT.md Section 4.1 (Operator, Integrator, Contractor).

2.2 Execution Pointers (The Toolchain)

Linguistic Precision: tools/lint/style.sh (Enforces no contractions).

Structure Verification: tools/verify.sh (Enforces Filing Doctrine).

Context Hygiene: tools/lint/context.sh (Enforces manifest compliance).

Truth Integrity: tools/lint/truth.sh (Enforces canon spelling).

DP Validation: tools/lint/dp.sh (Enforces TASK schema).

3. Current Dispatch Packet (DP)

DP-OPS-0046: Pointer-first agent constitution for Stela (no duplicated canon text).

3.1 Freshness Gate (Must Pass Before Work)

Base Branch: main

Required Work Branch: work/pointer-constitution-0046

Base HEAD: 46a04806

Gate Artifacts (Must Match):

- OPEN: storage/handoff/OPEN-main-46a04806.txt
- OPEN-PORCELAIN: storage/handoff/OPEN-PORCELAIN-main-46a04806.txt
- Dump: storage/dumps/dump-full-main-46a04806.txt
- Dump Manifest: storage/dumps/dump-full-main-46a04806.manifest.txt

Gate Commands (Must Pass):

- bash tools/lint/context.sh

3.2 Required Context Load (Read Before Doing Anything)

Loaded: PoT, SoP, TASK, CONTEXT, MAP

PoT.md (Jurisdiction, Git Authority, Behavioral Standard, Hard Constraints)

SoP.md (Recent DP patterns, verification expectations)

docs/MANUAL.md (Operator mechanics)

tools/lint/dp.sh (DP lint and TASK schema enforcement)

3.3 Scope and Safety

Objective: Finalize the Pointer-First Agent Constitution by refining TASK.md to remove duplicated canon text and refreshing the system context.

Constraints:

No scope expansion beyond the allowlist.

No new directories.

Stop on any lint failure until resolved within scope.

Target Files allowlist (hard gate):

TASK.md

SoP.md

storage/handoff/DP-OPS-0046-RESULTS.md

llms.txt

llms-small.txt

llms-full.txt

llms-ops.txt

llms-governance.txt

3.4 Execution Plan (A-E Canon)

3.4.1 State

Current Status: TASK.md contains legacy structure or "Ghost Canon" (duplicated rules).

Target State: TASK.md fully aligned with Pointer-First Constitution (PoT.md referenced, not copied). Context bundles refreshed.

3.4.2 Request

1. Refine TASK.md to strictly follow the Pointer-First schema.
2. Verify TASK.md against tools/lint/dp.sh.
3. Refresh context bundles via ops/bin/llms (Addendum requirement).
4. Capture proofs in RESULTS.

3.4.3 Changelog

Refined TASK.md to Pointer-First schema.

Refreshed llms context bundles.

3.4.4 Patch / Diff

(Applied directly).

3.4.5 Receipt (Required)

Populate: storage/handoff/DP-OPS-0046-RESULTS.md with:

Status: Scope summary, Tracked change, Verification.

Changelog: Human-readable summary.

Freshness Gate: Raw output of git rev-parse and git status.

Platform Dump: Path and size check.

Verification: Raw output of all linter commands (style, library, context, truth, verify, dp).

Diff Summary: git diff --stat output.

Mandatory Closing Block: (See Section 4).

4. Closeout (Mandatory)

Update TASK.md: Set Current Dispatch Packet to the next ID or placeholder.

Update SoP.md: Append a log entry describing the constitution refinement.

Verify: No off-allowlist edits (validate via git diff and tools/verify.sh).

Mandatory Closing Block

Varied Wording provision: Each entry must use meaningfully distinct wording; copy or minor tense changes are not acceptable.

Entry 4 must differ from Entry 1.

Primary Commit Header (plaintext)

Pull Request Title (plaintext)

Pull Request Description (markdown)

Final Squash Stub (plaintext) (Must differ from #1)

Extended Technical Manifest (plaintext)

Review Conversation Starter (markdown)

4.1 Thread Transition (Reset / Archive Rule)

Append a THREAD END entry to the TASK.md Work Log at completion.

Ensure the next session begins with a fresh OPEN artifact.

5. Work Log (Timestamped Continuity)

2026-02-10 19:00 - THREAD START: DP-OPS-0046. Seed: Pointer-first agent constitution refinement.

Works cited

1. dump-full-main-46a04806.manifest.txt